

ICS 604: Applied Data Science – Final Project Proposal

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Motivation

Prediction markets like Kalshi have emerged as a legal alternative to traditional sports betting in states like Hawaii. Unlike conventional sports betting, these markets frame outcomes as binary yes/no propositions about future events, providing a unique opportunity to study how market sentiment aligns with statistical reality.

NBA markets are particularly interesting to study due to significant market size disparities that may introduce exploitable biases. For example, the Los Angeles Lakers dominate media coverage as a large-market team featuring historical star LeBron James and newly acquired international sensation Luka Dončić. This media attention may inflate market prices beyond what underlying statistics would support. What initially drew me to exploring sentiment versus reality was the recent NFL playoffs, where the Los Angeles Rams received disproportionate media coverage in the NFC Championship despite losing to the eventual Super Bowl champions, the Seattle Seahawks. This suggested that media sentiment can meaningfully influence market pricing in ways that diverge from actual outcome probabilities.

The binary nature of these markets—resolving definitively to yes or no—makes them ideal for applying hypothesis testing, probability distributions, and regression techniques covered in this course. I expect to identify systematic biases in market pricing and demonstrate whether statistical models can form the foundation of a winning strategy.

Dataset

Primary data source: The Becker (2026) Kalshi dataset, publicly available on GitHub, contains 72 million trades including comprehensive NBA market data from 2025. This dataset provides market prices, trading volumes, timestamps, and outcomes for resolved markets. I will filter to NBA games from January-October 2025, extracting market-implied probabilities and actual outcomes for analysis.

External data sources: Basketball-Reference.com will supply team statistics including net rating (points per 100 possessions differential), win percentages, and recent performance trends. The GDELT Project database will track news article mentions and sentiment scores for each team in the 48 hours preceding games.

Scope: The analysis will focus on NBA games from the 2025 season with corresponding Kalshi markets. Based on the dataset's coverage, I expect to analyze 1,000-3,000 games, providing substantial statistical power.

Key variables: Market-implied probabilities, actual outcomes, team net rating differentials, and media attention metrics.

Anticipated challenges: GDELT data requires filtering to isolate sports coverage. Media coverage may be partially endogenous to team performance (successful teams

receive more coverage), requiring careful interpretation of results. While complete causal identification is beyond this project's scope, I will assess whether media effects persist after controlling for objective team quality metrics.

Research Questions

Primary Research Question: Do NBA prediction markets exhibit systematic pricing bias correlated with media attention disparities?

I hypothesize that teams receiving disproportionate media coverage are systematically overpriced in Kalshi markets relative to their actual win probabilities, representing a 'big market bias' effect independent of objective team quality. To test this, I will:

1. Quantify media attention disparity using GDELT news mention data for each team in the 48 hours preceding games
2. Calculate pricing errors (market-implied probability minus actual outcome) for each game
3. Use regression analysis to test whether media disparity predicts pricing error after controlling for team quality, home court advantage, and recent performance
4. Conduct hypothesis testing to compare market calibration in high versus low media disparity games

Optional Extension (if time permits): If systematic bias is detected, explore whether incorporating media disparity metrics into a logistic regression model improves prediction accuracy beyond using team statistics alone.

Methods / Approach

Data Collection: Historical NBA game data will be extracted from the Becker (2026) Kalshi dataset, including market prices and final outcomes. Team performance statistics will be collected from Basketball-Reference. Media attention will be quantified using the GDELT Project API to extract article mentions and sentiment scores for each team in the 48-hour window preceding games.

Feature Engineering: A composite **media disparity score** will be calculated as the simple average of two normalized components: (1) mention volume ratio between teams and (2) sentiment differential. This approach balances simplicity with capturing both attention and tone dimensions. Additional variables include pricing error, team net rating gap, home court advantage, and recent form.

Statistical Analysis: The primary analysis will employ **linear regression** (Week 12 material) to model pricing error as a function of the media disparity score while controlling for confounding factors. **Bootstrap resampling** (Week 8 material) will be used to construct confidence intervals for regression coefficients and test statistical significance of the media bias effect. Games will be stratified by media disparity levels to examine if high-disparity markets exhibit different pricing patterns.

Time Series Analysis: Time series methods (Week 14 material) will examine how Kalshi market prices evolve in the days leading up to games and whether price movements correlate with changes in media coverage.

Course Alignment: This approach applies regression modeling, bootstrap-based hypothesis testing, and time series analysis from Weeks 8, 12-14.

References

Becker, J. (2026, January 18). *The microstructure of wealth transfer in prediction markets*. <https://www.jbecker.dev/research/prediction-market-microstructure>